

On the Systematic Design Framework for Intuitive Interfaces in Telerobotic Operations: A Case Study in Surgical Robotics

Serin Cha^{1†}, Yoonsue Choi^{1†}, Hyeonji Bae², Chaewon Hwang³, Seok Chang Ryu^{1*}

Abstract—This paper proposes a design framework for intuitive interfaces in telerobotic operation, moving beyond outcome-based evaluation toward systematic interface design. A human-centered framework is introduced to guide interface strategy selection based on learning dynamics and to support implementation-level design decisions through analysis of human sensorimotor interaction. As a case study, the framework is instantiated through an embodiment-oriented index finger interface for controlling a distally bendable surgical instrument with unconventional kinematics. A telerobotic system was implemented, and the finger interface was compared with a familiarity-driven stylus interface in a dual-session human-subject study consisting of a training session and a retention session two weeks later. Results reveal a distinction between initial and later-stage interaction characteristics. While the stylus interface exhibited higher initial performance, the finger interface showed faster learning progression and achieved higher performance after learning stabilized. These findings suggest that embodiment-oriented interface design can support intuitive control by facilitating sensorimotor adaptation beyond immediate familiarity. Although evaluated in a surgical context, the proposed framework is applicable to the design of teleoperation interfaces for robotic systems with unconventional kinematics in broader domains.

Index Terms—Haptics and haptic interfaces, telerobotics and teleoperation, medical robots and systems, human-centered design.

I. INTRODUCTION

TELEROBOTIC operation has rapidly advanced as an enabling technology for human augmentation in high-risk environments such as surgery, space, and disaster response [1]. Within this broad domain, the present study focuses on teleoperation tasks that require continuous motion control, where operators modulate movements in real time rather than issue discrete commands [2]. Although significant progress has been made in mitigating technical challenges such as communication delay, sensing limitations, and control robustness, far less attention has been given to how effectively human operators can perform such control tasks [3]. Recent reviews

indicate that robotic surgical platforms have become increasingly diverse, both in their morphological structure and in the development of more articulated and flexible end-effectors [4], [5]. Broader analyses further show that ongoing technological advances have increased the functional complexity of these systems, posing greater challenges for reliable manual control under conventional teleoperation paradigms [6], [7].

Against this backdrop, prevailing paradigms for defining intuitiveness do not fully accommodate the diversity of modern robotic systems [8]–[10]. In human-computer interface (HCI) and human-robot interaction (HRI) research, intuitiveness is often characterized as relying on prior knowledge and being performed quickly and largely non-consciously [11]. Such an interpretation becomes increasingly limited as robotic mechanisms depart from familiar forms of interaction—particularly when their kinematics lack clear sensorimotor correspondences with everyday human actions.

This challenge becomes especially pronounced in emerging robotic platforms whose kinematics diverge substantially from rigid-body assumptions, making conventional notions of operator–robot correspondence inadequate. Continuum and bendable robots exhibit kinematic behaviors that differ fundamentally from those of conventional rigid tools [12]–[14]. Their nonlinear, high-degree-of-freedom, and continuously deformable structures create interaction dynamics that lack clear counterparts in everyday human experience, thereby exposing a key limitation of prior knowledge-based notions of intuitiveness. Addressing this mismatch between human interaction patterns and robotic kinematics, prior studies to explore interface design approaches that either leverage widely adopted input devices to capitalize on users' prior experience [15]–[17], or reproduce the robot's morphology or kinematic structure through dedicated control mechanisms [18]–[20].

While these approaches provide valuable design insights for particular robotic mechanisms, they are often evaluated with an emphasis on early-stage performance before meaningful adaptation has occurred and tend to focus on optimizing specific interface properties within a given task or robot configuration. In some cases, this evaluation perspective may limit the ability to capture how interface design supports sustained interaction and learning over time. As a result, they offer limited guidance for systematically selecting and implementing interface design strategies across different robotic morphologies and task contexts [21], [22].

Accordingly, this study introduces a structured framework for intuitive teleoperation interface design in which learning

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dynamics are treated as an explicit design consideration. Rather than focusing solely on which interface yields superior initial usability, the framework examines how different design strategies support early adaptation, learning progression, and stabilized performance over repeated interaction.

The framework is structured around two key questions:

- 1) *Which interface design strategy is appropriate when learning dynamics are considered?*
- 2) *How can the selected strategy be systematically implemented at the system level under task- and robot-specific constraints?*

The first question is addressed by formulating teleoperation as a linkage structure connecting the Human, the Interface, and the Robot (**Fig. 1(a)**). This formulation enables prior interface approaches to be interpreted according to which part of the teleoperation linkage they primarily reinforce. Familiarity-based approaches [15]–[17] primarily strengthen the Human–Interface linkage by leveraging practiced motor patterns (**Fig. 1(a-i)**), whereas kinematic-similarity approaches [18]–[20] more directly support the Interface–Robot linkage by aligning the motion structure of the interface with that of the robot (**Fig. 1(a-ii)**). When the follower robot resembles tools or motion patterns already familiar to the operator, these strategies may act in a complementary manner.

In contrast, for continuum or bendable robots with unfamiliar morphology and motion, these strategies often impose competing demands. Increasing kinematic fidelity to the robot may require motion patterns that feel unnatural to the operator, whereas preserving familiar interaction patterns can introduce structural mismatches that hinder precise manipulation.

These trade-offs become more evident when the teleoperation linkage is considered from the operator's standpoint. From this perspective, the problem is not simply how to optimize isolated mappings within the chain, but how to establish a coherent Human–Robot linkage across the overall sensorimotor loop. This observation motivates an *embodiment-oriented* design strategy, which seeks to reinforce the Human–Robot linkage at the level of the full interaction loop rather than at individual stages of control mapping (**Fig. 1(a-iii)**).

The sense of embodiment—often described as a sensory illusion in which an artificial system is experienced as part of one's own body—has been extensively studied in haptics, telepresence, and human augmentation [23]. Prior findings associate embodiment with enhanced bodily awareness, more natural motor engagement, and improved dexterity, suggesting that embodiment-related sensorimotor coherence can support faster skill acquisition, stable fine manipulation beyond initial learning, and performance improvements over time, even when interacting with unconventional robotic mechanisms [24]. From a learning-dynamics perspective, these findings suggest that embodiment-oriented strategies may offer advantages that are not fully captured by initial task performance alone, but instead emerge through repeated interaction and sustained use.

The second question then concerns how such an embodiment-oriented strategy can be translated into implementable design conditions. While prior embodiment-related studies have primarily examined embodiment as a perceptual or experiential phenomenon [23], [25], often emphasizing

factors such as sense of agency or visuomotor congruency, the present framework extends this discussion into a system-level design formulation. In particular, teleoperation is reinterpreted in human-centered terms as a structured sensorimotor interaction among the human, the interface, and the robot, organized around the concept of the *Perceived Body*—a task-relevant representation of the robot as experienced by the operator. On this basis, embodiment-related principles are translated into concrete interface- and system-level design considerations grounded in the interaction mechanisms among these components.

Surgical teleoperation serves as a representative application domain for instantiating and empirically evaluating the proposed design framework. In minimally invasive surgery, surgeons must perform delicate manipulation through long instruments while relying on mediated visual feedback, making intuitive control particularly critical. At the same time, emerging surgical robots increasingly incorporate flexible and articulated mechanisms, which amplify the challenges associated with establishing intuitive operator–robot interaction.

This paper presents the following contributions:

- A human-centered framework for teleoperation interface design that treats intuitiveness not as a single property but as a design choice among alternative strategies, and incorporates learning dynamics as a criterion for strategy selection.
- An operationalization of an embodiment-oriented design strategy into implementable interface-level considerations centered on the *Perceived Body*, demonstrated through a proposed index-finger interface for distally bendable instruments.
- A dual-session human-subject case study in surgical robotics that provides empirical support for the proposed framework by comparing two interface strategies across learning phases, revealing distinct strengths in initial performance and learning progression.

The remainder of this paper is organized around the proposed design framework and its realization and evaluation. Section II introduces the human-centered design framework, covering strategy selection based on learning dynamics and embodiment-oriented design principles grounded in human sensorimotor interactions. Sections III and IV describe its realization through the finger-based interface, teleoperation system, and stylus-based baseline interface. Sections V and VI then present the dual-session human-subject experiment and its results, including an analysis of representative movement strategies. Finally, Section VII concludes the paper by summarizing the key findings and discussing their implications, limitations, and directions for future work in intuitive teleoperation interface design.

II. HUMAN-CENTERED DESIGN FRAMEWORK

The framework is organized into two levels: (A) strategy selection, which determines appropriate interface design strategies based on learning dynamics, and (B) design implementation, which translates the selected strategy into concrete system- and interface-level design considerations. In addition,

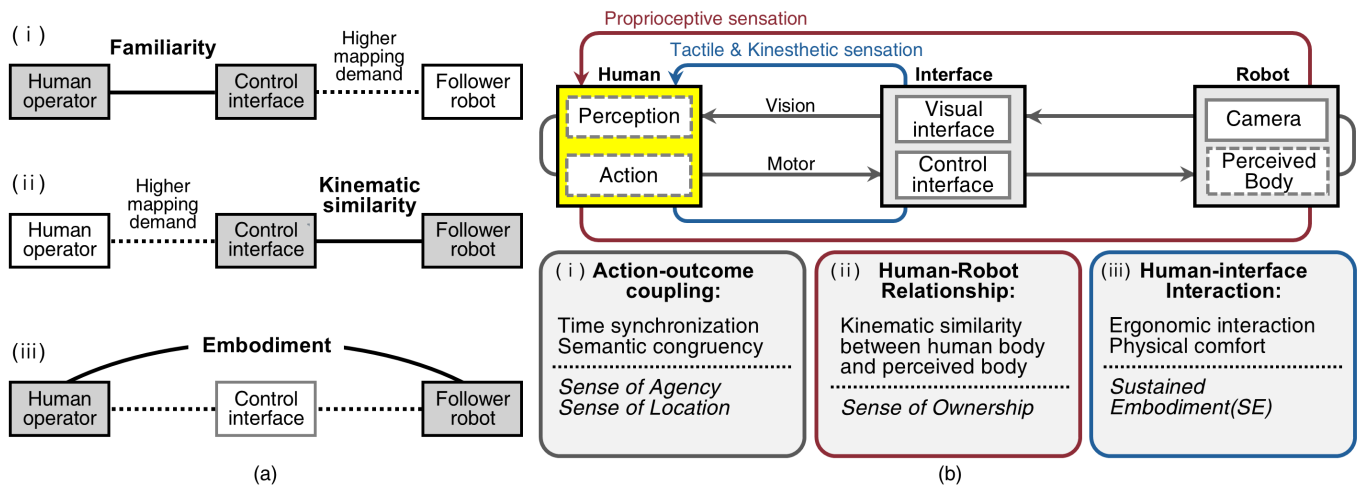


Fig. 1. Conceptual illustration of the proposed human-centered design framework organized into two levels. (a) Strategy selection level, illustrating three interface design approaches within the teleoperation interaction structure; (i) Familiarity-based strategy employs input devices that are already familiar to the operator and primarily reinforce the Human–Interface linkage by leveraging practiced motor patterns. (ii) Kinematic-similarity strategy adopts robot-mimicking interfaces that emphasize the Interface–Robot linkage by aligning the motion structure of the interface with that of the robot. (iii) The proposed embodiment-based strategy aims to strengthen the Human–Robot linkage across the entire sensorimotor loop. (b) Design implementation level, organizing interaction-based design considerations around three human-centered interaction mechanisms; (i) Action–outcome coupling, addressing temporal synchronization and semantic congruency to support the senses of agency and location. (ii) Human–Robot relationship, focusing on kinematic similarity between the human body and the perceived body to support the sense of ownership. (iii) Human–Interface interaction, emphasizing ergonomic interaction and physical comfort to sustain embodiment during continuous operation.

this section clarifies how the proposed framework is instantiated in the present case study.

A. Strategy selection

Familiarity-based strategies (**Fig. 1(a)-i**) aim to achieve intuitiveness by leveraging prior experience and established motor skills. Consistent with common definitions of intuitiveness in the literature [11], interfaces are often considered intuitive when they allow users to rely on prior knowledge and perform effectively with minimal initial training [26]. As a result, evaluation of intuitiveness has frequently emphasized initial usability or early task performance, favoring interfaces that adopt familiar input modalities or interaction metaphors.

However, this reliance on familiarity becomes problematic when teleoperating unconventional robotic systems. In such systems, particularly those requiring fine, continuous, and precise control, the mapping between a familiar control interface and an unfamiliar follower robot can become increasingly complex. When substantial kinematic dissimilarities exist between the human input space and the robot’s actuation space, operators may still achieve reasonable initial performance through prior motor habits. However, as interaction progresses, they may encounter limitations in performance stabilization or longer-term execution. This raises the question of whether maximizing initial usability alone is sufficient for selecting an intuitive interface strategy in such contexts.

In response to this limitation, learning dynamics is considered as an additional criterion for strategy selection [27], [28]. In this work, learning dynamics refers to observable changes in performance over repeated interaction, including changes in consistency, variability, and stabilization of control, rather than the underlying learning process itself or formal learning rates. From this perspective, intuitiveness is not treated as a

static property revealed at initial use, but as a time-dependent characteristic that may manifest differently across phases of interaction.

Embodiment-oriented strategies (**Fig. 1(a)-iii**) place greater emphasis on strengthening the functional coupling between the human operator and the robot, focusing on how sensorimotor coordination with the robotic system may develop through interaction [29], [30]. Rather than prioritizing familiarity at the interface level, these strategies aim to support interaction patterns that align with the operator’s motor organization and perceptual expectations, potentially facilitating more stable performance as experience accumulates. While familiarity-based and embodiment-oriented strategies are not mutually exclusive, their relative advantages are expected to depend on task demands and the stage of interaction.

A further line of work has explored kinematic similarity (**Fig. 1(a)-ii**) as an intuitive design approach by aligning the geometric structure of the interface with that of the robot. While such alignment can improve the transparency of how interface motions map to robot movements, it does not in itself specify how the operator should organize motor actions to achieve effective control. As a result, control strategies may remain implicit and user-dependent, leading to variability in operation across individuals and contexts. For this reason, kinematic similarity is not treated as an independent strategy in this paper, but rather as a supporting design consideration that may contribute to embodiment-oriented interaction depending on task context. In particular, kinematic similarity primarily concerns how interface motions are mapped to robot kinematics, rather than how operators organize motor behavior over learning.

Based on these considerations, this study adopts an embodiment-oriented strategy for intuitive teleoperation in the context of unconventional robots and precision-demanding

tasks. From a learning-dynamics perspective, this strategy is evaluated in comparison with a familiarity-based interface. The following subsection details the design implementation derived from embodiment-related cognitive and sensorimotor studies, which guide the realization of this strategy at the interface and system levels.

B. Design implementation

From a human-centered perspective, teleoperation can be understood as a set of coupled interactions centered on the operator (see **Fig. 1(b)**) [27]. At the core lies the primary action–outcome coupling of teleoperation, through which the operator’s motor actions give rise to perceived outcomes in the remote robot [30]. Beyond this core coupling, teleoperation involves interactions between the human and the robot, and between the human and the interface, each mediated by distinct sensorimotor signals. These interactions are shaped by multimodal sensorimotor cues, including vision, motor command, proprioceptive sensation, and, where available, tactile and kinesthetic information. The present framework therefore approaches design implementation as the organization of such cues across the human–interface–robot loop. Among these, haptic cues are especially relevant from an interface-design perspective because they can be directly shaped through the interface’s mechanical and sensing properties.

On this basis, embodiment-oriented interface design is examined by identifying how these interaction mechanisms can be shaped to support stable and intuitive teleoperation. The sense of embodiment is commonly understood to arise from three interrelated components—the senses of agency, ownership, and location—each of which relies on specific forms of sensorimotor congruency [31].

Beyond static perceptual paradigms such as the rubber hand illusion [32], teleoperation involves continuous movement and sustained interaction over time. In such settings, embodiment should not only be established, but also maintained throughout ongoing manipulation [29]. Conditions that disrupt embodiment—such as physical discomfort or interfering sensory noise—are therefore critical, as they can undermine stable control even after initial embodiment has been formed [30].

A human-centered account of teleoperation further requires distinguishing between the robot’s physical structure and the part of the robot that the operator perceives as being acted upon. In practice, operators do not interact directly with the mechanical components of the robot, but organize their actions around an internal representation of a Perceived Body (PB). Depending on task demands, kinematic structure, and available sensory cues, the PB may correspond to a specific segment or a broader functional region of the robot. This concept emphasizes the internal representation of the remote system by the human operator rather than its physical structure alone, and has been shown to play a critical role in embodiment and motor learning [29], [33].

(1) Action–Outcome Coupling (Fig. 1(b)-i): A fundamental interaction mechanism in teleoperation is action–outcome coupling, which describes how an operator’s actions give rise to behavior of the PB. Coherent action–outcome coupling

underlies the sense of agency, by supporting a clear perception of causality between action and effect, as well as the sense of location, by maintaining spatial coherence between motor action and visual feedback.

From a system-level perspective, supporting action–outcome coupling requires that the teleoperation pipeline preserves both temporal and semantic consistency between action and outcome. Temporal synchronization reduces delay and jitter, thereby helping maintain the immediacy of action–effect contingencies that underlies perceived causality and the sense of agency [27], [30]. Semantic congruency, in turn, preserves the intended meaning of motion—such as direction, plane, and functional role—across sensing, mapping, actuation, and visual feedback, thereby supporting sensorimotor congruence between the operator’s action and the behavior of the PB [23], [31].

These requirements translate into concrete design considerations across multiple layers of the teleoperation system. At the human–interface level, the interface should capture the operator’s intended motion without semantic distortion or loss of task-relevant degrees of freedom. At the interface–PB level, software mapping, robot actuation, and visual presentation should reproduce the captured motion semantics in a spatially coherent manner, including consistent alignment of coordinate frames, camera viewpoints, and reference axes.

(2) Human–Robot Relationship (Fig. 1(b)-ii): The human–robot relationship in teleoperation is closely tied to the sense of ownership over the PB, which depends on how coherently the operator can integrate the PB into their existing motor organization [29], [34]. From an interface design perspective, supporting this sense of ownership requires establishing kinematic similarity between the human body and the PB, such that control actions can be organized in a manner consistent with the operator’s proprioceptive expectations.

Kinematic similarity does not imply direct morphological correspondence between the human body and the robot. Rather, it refers to whether the structure and coordination of selected human input motions bear an organized resemblance to the controllable degrees of freedom of the PB. Accordingly, kinematic similarity functions as a relational property between human motion and robot behavior, rather than as a strict geometric constraint.

In practical interface design, this consideration translates into decisions regarding which human body motions should be employed for control. Such decisions are informed by prior analysis of the PB’s kinematic mechanism, the characteristics of the target task, and the extent to which different aspects of the PB should be actively controlled by the operator. Based on this analysis, candidate human body motions can be identified that exhibit structural similarity to the relevant PB behavior and are compatible with the operator’s motor organization.

(3) Human–Interface Interaction (Fig. 1(b)-iii): Even when coherent action–outcome coupling and kinematic similarity are achieved, embodied interaction must remain physically sustainable during continuous manipulation. At the action level, excessive joint excursions or muscular effort increase physical load and fatigue, while at the perceptual level, uncomfortable postures or sustained strain introduce sensorimotor

noise. Both factors can destabilize control and progressively erode embodiment over time.

Ergonomics therefore constitutes a critical interface design consideration for sustained embodiment, particularly in prolonged or high-precision tasks. At the action level, interface design should not be guided by kinematic similarity between the human body and the Perceived Body alone, but also by how comfortably the selected body part can perform the required control motions with minimal physical burden or fatigue. At the perceptual level, the interface should minimize sensory interference, discomfort, and physical constraints that may be perceived as noise, thereby avoiding disruptions to immersion and sustained embodied interaction.

Beyond ergonomics alone, human–interface interaction may also be shaped by the haptic properties of the interface itself. From an interface-design perspective, these may include passive mechanical support or guidance arising from the interface's own structure, as well as active haptic feedback when remote interaction forces are rendered to the operator. Depending on the task, such haptic cues may be expressed through kinesthetic/proprioceptive sensation, tactile contact, or both. When congruent with the selected control motions and the behavior of the PB, such cues can reduce sensorimotor ambiguity, support contact stability, and reinforce sustained embodied interaction. Conversely, incongruent mechanical or haptic cues may function as interference, increasing physical burden, perceptual noise, or instability during control.

Taken together, the interaction mechanisms discussed above define system-level conditions that must be satisfied for intuitive teleoperation, including coherent action–outcome coupling and stable perception of the PB. From the perspective of interface design, however, these conditions translate into a more focused set of design considerations centered on how the human operator physically and perceptually engages with the system.

Specifically, embodiment-oriented interface design requires deliberate decisions regarding which human body motions should be employed for control, guided by kinematic similarity between the human body and the PB, as well as by the physical sustainability of the required control actions. In addition, the interface must be mechanically designed to measure and transmit these motions in a manner that minimizes physical burden, perceptual discomfort, and sensory interference during sustained interaction, while also considering whether mechanical or haptic support can help scaffold stable control. Together, these considerations operationalize the embodiment-oriented strategy at the interface level, providing a concrete basis for translating the proposed framework into implementable interface designs.

C. Case study: surgical teleoperation

In surgical teleoperation, operators are required to generate smooth and continuous trajectories, rather than execute isolated or discrete actions, making time-dependent interaction characteristics particularly relevant for interface evaluation. Furthermore, distally bendable surgical instruments introduce kinematic relationships that do not directly correspond to conventional joint- or Cartesian-space control, requiring operators

to adapt to non-intuitive input–output mappings regardless of the chosen interface.

Following the interface-level design considerations of the proposed framework—particularly kinematic similarity between the human body and the PB, and ergonomic alignment with the operator's motor structure in terms of both action execution and perceptual comfort—an embodiment-oriented interface was designed and fabricated based on index finger motion.

In contrast, a stylus-based interface was adopted as a familiarity-driven baseline, as pen-like stylus interfaces are widely used in prior surgical and flexible instrument teleoperation studies and are likely to be familiar to users. Accordingly, the present comparison was designed to examine an embodiment-oriented interface strategy against a familiarity-oriented alternative grounded in prior motor experience.

Consistent with this comparative scope, the haptic aspect of the framework was instantiated only through passive mechanical support at the interface, while active environment-mediated haptic feedback was intentionally excluded in order to compare the two interface strategies under matched control conditions. For the present follower system, a dedicated robot-mimicking baseline was also not straightforward to define, because the instrument combines distal bending, rolling, and translation under an RCM constraint, such that the kinematic structure to be mirrored on the leader side is not uniquely determined and would require additional design decisions regarding which motions should be reproduced and how they should be coordinated.

While communication delay and operator workload are fundamental considerations in teleoperation systems and are addressed as necessary conditions in the proposed design framework, the surgical context allows for relatively greater tolerance to modest delays due to the emphasis on deliberate, precise motion [28]. The present case study therefore focuses on how interface design strategies influence time-dependent performance characteristics within a fixed experimental platform.

III. INDEX FINGER INTERFACE

A. Index finger for the bendable distal section

An evaluation of kinematic similarity begins with an analysis of the follower robot's kinematics, with particular attention to the region perceived and manipulated by the operator as the Perceived Body. The instrument comprises a single bendable distal section and provides full three-dimensional workspace coverage through distal bending, axial rolling, and translational motion of the instrument shaft. A reference point P is defined at the base (proximal end) of the bending section (**Fig. 2(a)**). This motion set was adopted as a compact configuration for achieving spatial manipulation in constrained environments.

A corresponding motion capability exists in the human body—specifically, the finger (**Fig. 2(b)**). The finger consists of the metacarpophalangeal (MCP), proximal interphalangeal (PIP), and distal interphalangeal (DIP) joints, whose coordinated flexion enables a bending motion analogous to that of the instrument's distal section. Axial rolling of the instrument

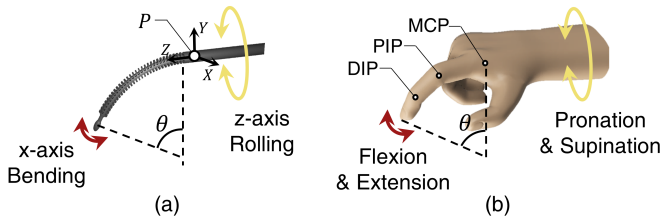


Fig. 2. Kinematic similarity between human index finger and surgical instrument with bendable distal section: (a), (b)

corresponds to pronation and supination of the wrist, while the distal location of both the bendable section and the finger supports coherent spatial perception of tool position. An additional advantage arises from the natural decoupling between global and local motion. Global positioning is performed by the arm and hand, whereas local shape modulation is achieved through finger flexion, allowing independent control of the distal section. Moreover, the fingertip contains a high density of mechanoreceptors [35], making it particularly suitable for transmitting force-related cues from the tool tip.

Before selecting the finger as the target control body part, alternative candidates were examined in terms of kinematic similarity and ergonomic feasibility. Wrist motion has been proposed as a kinematically similar strategy for controlling bendable sections [36]. However, as the length of the bending section increases, achieving comparable distal curvature would require relatively large wrist excursions, increasing joint displacement and physical strain; this makes wrist-based control less suitable for prolonged or precise manipulation. The elbow, although part of a continuous kinematic chain with the shoulder and wrist, is likewise less appropriate as a primary control source because precision tasks benefit from stable elbow support. In contrast, finger flexion and extension are central to everyday fine manipulation [37]. Among all fingers, the index finger is anatomically and neurologically the most independent [38], and its established roles in pointing and precision motor control [39]–[41] make it particularly suitable for sustained and accurate manipulation.

Considering kinematic similarity together with ergonomic feasibility, the index finger was selected as the target body part for implementing the proposed embodiment-oriented design considerations. On this basis, a control interface was designed to allow operators to perceive and manipulate the bendable distal section as an extension of their own index finger.

B. Design features of the index finger interface

Once index-finger motion was selected as the control source, the interface had to measure finger flexion and global hand motion without interfering with natural movement, while also supporting comfortable and stable fine control during prolonged use. Purely gesture-based or optical tracking approaches were therefore not adopted, because unrestricted motion alone provides limited contact stability and limited passive haptic/proprioceptive support for precision teleoperation. Instead, the present design sought physically grounded interaction at the interface so that sensing, comfort, and mechanical support could be integrated to stabilize ongoing control. Such

physical grounding is beneficial not only for providing passive control-relevant cues during prolonged manipulation, but also because it offers a more suitable basis for future incorporation of active haptic feedback, where rendered forces must be coupled coherently to the user's control motions through the interface.

The proposed finger interface (**Fig. 3(a)**) consists of two primary components: a finger flexion measurement unit (**Fig. 3(b)**) and a handle (**Fig. 3(c)**). The finger flexion unit measures the flexion angle of the index finger, while the handle is connected to a six-degree-of-freedom motion platform to capture global hand movement. During operation, the user holds the handle and places the distal phalanx of the index finger into a thimble attached to the flexion measurement unit.

The finger flexion measurement unit comprises a three-degree-of-freedom serial mechanism for measuring joint flexion, together with a ball joint that provides redundant degrees of freedom to accommodate comfort-related MCP motion. Hinge joints 1 and 2 incorporate torsion springs that generate forces pushing the fingertip inward and lifting it during flexion. This configuration provides passive mechanical support at the fingertip and modulates the interaction impedance, improving stability and precision while preventing finger slippage without additional fixation. In addition, the spring stiffness at Joint 1 is set higher than that at Joint 2, thereby allowing only limited MCP joint rotation while passively discouraging excessive MCP-dominant flexion. This passive mechanical bias encourages motion patterns that are more consistent with the bending behavior of the instrument. Hinge joint 3 is aligned with the center of the DIP joint, ensuring synchronous motion of the thimble and distal phalanx.

The handle is integrated with the motion platform to measure the position and orientation of the hand, corresponding to the base of the instrument. As shown in **Fig. 3(c)**, the handle is tilted by 30° to reflect anatomical considerations. In a neutral wrist posture, the index finger is naturally offset from the axial rotation axis; the handle tilt aligns the finger with this axis, supporting intuitive control. In addition, the diagonal fold formed by the thenar eminence and thumb web during gripping enhances grip stability and force transmission [42].

The axial rotation axis of the handle aligns with the anatomical center of forearm pronation and supination, which in the grasping posture lies approximately along the line of the ring finger. When the hand and forearm are treated as a single linkage to minimize wrist motion, their rotations become interdependent. Aligning the handle's rotational axis with this anatomical reference enables comfortable and intuitive control of axial rotation without additional cognitive effort.

The prototype values used in the present study are reported in Section IV. These parameters may be adjusted in other implementations to accommodate user- or application-specific requirements without changing the core design rationale.

IV. EXPERIMENTAL SYSTEM SETUP

A telerobotic system was developed to enable human-participant evaluation of the finger interface, with a stylus

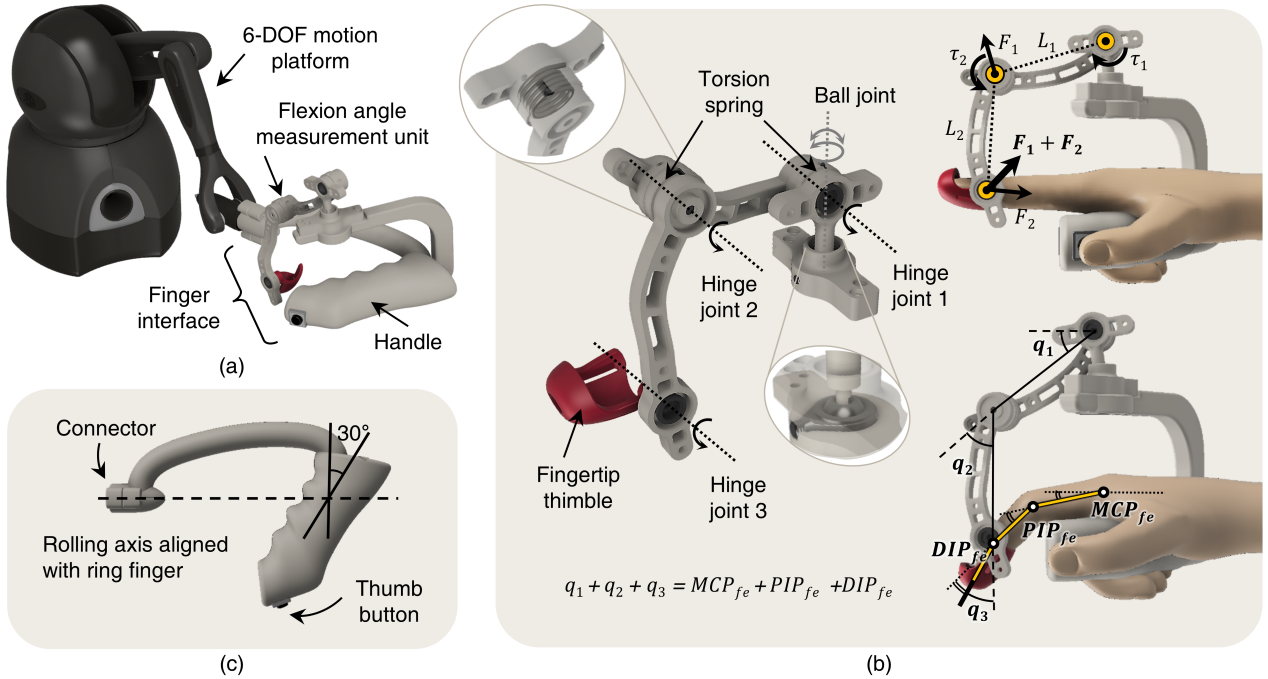


Fig. 3. Design of the finger interface, including its connection to the motion platform and individual components: (a) Example of the finger interface connected to a 6-DOF motion platform, (b) Flexion measurement of the index finger, (c) Ergonomic handle with an optional auxiliary thumb-operated button (not used in the present experiment).

interface used as a comparative condition. The system allows participants to perform remote manipulation tasks using a distally bendable instrument, and the overall system architecture is illustrated in **Fig. 4(a)**.

The system consists of a leader system (**Fig. 4(b)**) and a follower system (**Fig. 4(c)**). The leader system enables user interaction through either the stylus interface (**Fig. 4(d)**) or the finger interface (**Fig. 4(e)**), both of which are mounted on a 6-DOF motion platform for capturing translational and rotational inputs. Interface-specific motion alignment was implemented through customized mapping algorithms implemented at the system level, as detailed in Section IV-C.

The follower system comprises a 6-DOF robotic manipulator and a custom actuation module. The manipulator executes translational and rolling motions under a Remote Center of Motion (RCM) constraint, reflecting the constrained entry conditions typical of minimally invasive surgery. The actuation module, mounted at the end-effector, controls the 1-DOF bending section of the instrument (**Fig. 4(f)**). A foot clutch was integrated to allow participants to reposition the input interface when reaching the workspace limits of the motion platform.

The centralized server operated with an update period of approximately 100 ms, encompassing the full control loop from leader-side input acquisition through mapping, command execution, and state feedback logging. Based on this, the application-level timing of the system loop was characterized using client-side timestamps as the time difference between command transmission and reception of the corresponding state feedback (100.23 ms). Although this value does not represent the full human-perceived interaction latency, it reflects

the system-side control and communication timing determined by the server update cycle. Visual feedback latency was minimized by directly connecting the camera to the display system via a wired interface, such that the remaining delay was dominated by typical camera acquisition and display latency.

Prior studies have shown that system delays on the order of 100–200 ms are generally tolerable for human operators in teleoperation scenarios [43]. Performance degradation in simulated robotic surgical tasks was also reported to be limited for delays below approximately 300 ms, with substantial difficulty emerging at higher latencies [44]. In the present study, both interfaces were evaluated under identical system configurations, including server update timing and visual feedback pathways. Accordingly, the present findings should be interpreted as a comparison between interface strategies within a fixed teleoperation platform configuration, rather than as evidence that the observed differences are invariant to changes in control bandwidth. Sensitivity to higher-frequency update conditions was not separately evaluated in the present study.

A. Prototypes of the finger and stylus interfaces

The finger interface prototype employed link lengths of $L_1 = 56$ mm and $L_2 = 61$ mm, as labeled in Fig. 3, for the proximal and distal segments, respectively. Hinge Joints 1 and 2 used torsion springs with estimated angular stiffnesses of approximately 1.97 N·m/rad and 1.18 N·m/rad, respectively. Angular displacements at Hinge Joints 1–3 of the finger interface were measured using 14-bit magnetic encoders (SAE360API-12DN-XY, SERA, Republic of Korea), interfaced to a microcontroller (Arduino UNO, ARDUINO,

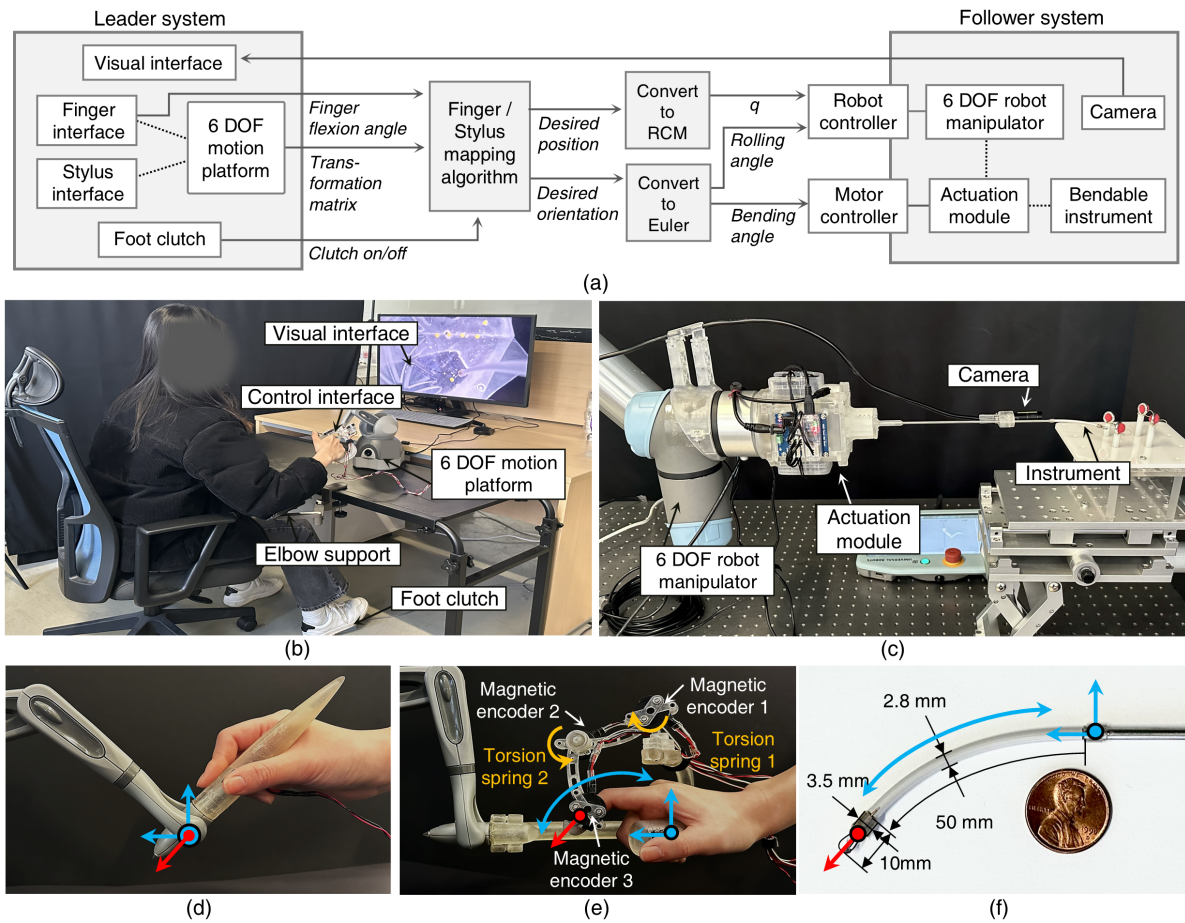


Fig. 4. Overview of the experimental system: (a) Block diagram illustrating the control flow and communication between the leader and follower systems, including motion mapping and control algorithms, (b) Leader systems, (c) Follower systems, (d) Prototype of the stylus interface, (e) Prototype of the finger interface, (f) Instrument with a bendable distal section.

Italy) via 10-bit ADC, yielding an effective resolution of approximately 0.5° after accounting for quantization and noise. Both the finger and stylus interfaces were fabricated using clear resin (Clear V4) with a Form3 3D printer, ensuring consistency in material and manufacturing. The stylus replica was 3D-modeled to match the geometry of the commercial version for fair comparison. The total weight of the finger interface prototype was 215 g, whereas the stylus interface weighed 19.7 g.

B. Instrument and actuation module

The instrument comprises a composite structure, with a distal section made from a laser-machined NiTi tube (JMM19841, Johnson Matthey, United Kingdom, OD/ID: 1.95/1.65 mm) and a proximal section from stainless steel 304 tubing (OD/ID: 2.8/2.5 mm) [45]. Laser machining was performed using femtosecond laser technology (Starcut Tube SL, Coherent, USA) at the Glocal Interventional Mechano-Biotechnology Convergence Research Center (Jeonju, Republic of Korea). The instrument was designed based on a mechanical model that accounts for friction during bending, enabling a constant-curvature bending angle of up to 90° . For system stability, the operating range of the motor was restricted such that the instrument would not exceed a maximum bending angle

of 90° . The bending motion is actuated via a tendon-driven mechanism using Kevlar tendons with minimal elongation, guided through a multi-lumen tube (Pebax5533SA01MED, ARKEMA, France). The instrument is detachable from the actuation module (Fig. 4(c)), which uses a DC motor (AX-12A, ROBOTIS, Republic of Korea) to drive the tendons. The assembly, integrated with a robotic arm, provides bending via the module and translation and rolling via the robot for full 3D manipulation.

C. Position and orientation mapping

Both interfaces operated within a common task reference frame to ensure consistent visuo-motor correspondence between operator input and instrument motion. Translational and rotational commands were generated through interface-specific mappings while preserving directional and planar consistency in the perceived instrument behavior.

Translational input for the finger interface was computed from handle displacement, using a modified transformation matrix of the motion platform, whereas the stylus interface employed the platform's default position reference. After translational inputs from both interfaces were computed, a common motion-scaling ratio of 20:1 was applied to all

translational commands to scale operator movements to the instrument workspace for safe and precise manipulation.

In both interfaces, the instrument tip orientation was realized through a combination of bending and rolling motions, while the mapping from user input to these motions differed according to interface characteristics. For the finger interface, the finger flexion angle was mapped one-to-one to the bending angle of the instrument's distal section, normalized to the instrument's maximum bending range of $\pm 90^\circ$. Hand supination and pronation were independently mapped to instrument rolling, resulting in a decoupled control structure.

The stylus-tip orientation, defined relative to its base frame, was represented as a rotation matrix. It was then mapped onto the two rotational degrees of freedom available at the instrument tip, namely bending and rolling. The rotational component that could not be realized by the instrument's two-DOF tip kinematics was excluded. Both bending and rolling angles were constrained within $\pm 90^\circ$, and the desired tip orientation was approximated through an appropriate combination of these two motions. This mapping was chosen to preserve the stylus as a familiarity-oriented, pen-like orientation controller, rather than introducing a separate dedicated bending input that would alter the conceptual nature of the baseline interface.

At the system level, the robot executed the mapped translational and rolling commands, while the bending angle was sent directly to the instrument actuator. The mathematical formulation of the orientation decomposition and its projection onto the instrument's kinematic subspace is provided in the supplementary material.

Additional implementation details, data, and materials related to the mapping and experimental tasks are available from the authors upon reasonable request.

V. DUAL-SESSION EXPERIMENTAL DESIGN

Interface intuitiveness was examined through a dual-session human-participant experiment designed to evaluate performance across different stages of interaction. Rather than focusing solely on initial task execution, the experimental design captured performance during early interaction, after structured training, and after a temporal gap intended to reflect stabilized interaction performance.

A. Task environment

Three types of tasks were conducted in total: two training tasks and one bladder mock-up task designed to reflect the actual size of a human bladder (**Fig. 5(a)**). The first training task (Task 1) involved reaching four target positions and was designed to be achievable using only translational motion. However, combining rolling and bending motions can reduce the required movement distance, enabling faster task completion. The second training task (Task 2) required the instrument to maintain its position while passing through a ring and reach two targets located to the left and right relative to the insertion path. This constraint required coordinated use of rolling and bending motions.

The bladder mock-up task involved reaching and removing multiple targets inside a bladder model, simulating cancerous

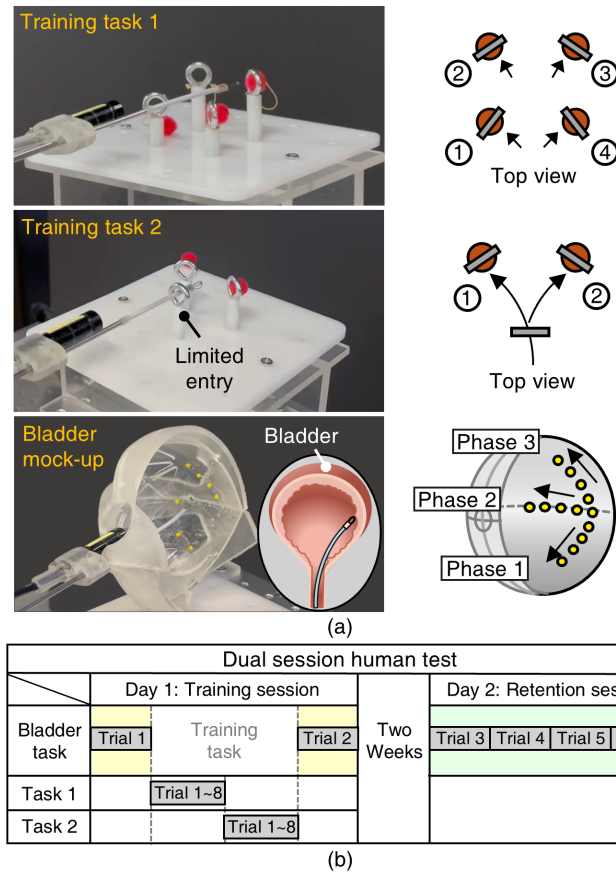


Fig. 5. Task environment and test protocol: (a) Overview of the simulated bladder surgery task and two training tasks: the left panels display the experimental setup, and the right panels illustrate the step-by-step task sequence. (b) Dual-session test protocol for human subjects.

lesions representative of those treated in cystectomy. Tasks requiring broad accessibility through a narrow entry, such as those inside the bladder, are well suited for small bendable instruments. This task was therefore selected as a suitable scenario to compare interface performance for controlling such instruments. The bladder mock-up task consisted of removing targets following curved paths from three distinct locations: lower, middle, and upper positions, corresponding to Phase 1, Phase 2, and Phase 3, respectively.

Training task environments were built using a commercial laparoscopic kit, with 10 mm-diameter soft targets spaced 37 mm apart. The bladder model was designed as an ellipsoid (125 mm $\times$ 90 mm), reflecting real human bladder dimensions [46]. Small targets (2 mm in diameter) were used in the bladder mock-up task to increase difficulty and require higher precision, while larger targets were used in training tasks to support learning.

All tasks were performed in a predetermined order, and subjects were instructed to follow this sequence exactly. In Training Task 1, subjects were instructed to remove four targets sequentially in a predetermined clockwise order (**Fig. 5(a)**). In Training Task 2, subjects were instructed to first remove the target on the left and then the one on the right. A target was considered successfully removed when the small red spherical element embedded in the ring-shaped structure

became detached.

B. Experimental protocol

The human-subject test was divided into two sessions, Day 1 and Day 2, separated by a two-week interval to assess the stability of performance after training (**Fig. 5(b)**). On Day 1, subjects first performed a baseline trial of the bladder mock-up task to establish their initial performance level. This was followed by two structured training tasks—Training Task 1 and Training Task 2—each repeated eight times to facilitate learning. Upon completion of the training phase, subjects performed a second trial to assess improvement after training. After completing the full protocol with the first interface, which was randomly assigned, subjects took a 15-minute break before repeating the same procedure with the second interface. Two weeks later, on Day 2, subjects performed the bladder mock-up task four times with each interface. The order of interfaces was reversed from Day 1 to minimize any potential order effects. Because Day 1 employed a within-subject crossover design, the 15-minute break and reversed interface order on Day 2 were intended to mitigate, but not fully eliminate, possible order or carryover effects. After completing all tasks, they were asked to fill out the NASA-TLX questionnaire for each interface.

C. Participant demographics

A total of 11 right-handed adult subjects participated in the experiment; one subject was excluded due to protocol deviation. The remaining 10 participants (5 females, 5 males; age 19–27) were included in the analysis. The study was approved by the Institutional Review Board with reference number ewha-202206-0031-06. All participants were informed of the purpose and procedures of the study prior to participation and provided written informed consent.

Participants with no prior experience using medical or telerobotic interfaces were recruited, as the study focuses on how interface design shapes intuitiveness across different learning phases rather than on expert-level performance. This choice allows performance differences arising from learning efficiency, performance stability over time, and stabilized interaction to be examined more directly, without being masked by pre-existing control strategies.

D. Evaluation metrics and statistical methods

Intuitiveness was evaluated using a combination of objective task-performance metrics and subjective workload assessment, capturing both how efficiently the interfaces were used and how they were experienced by the operators.

Task completion time (T_{TC}) and total tip trajectory distance (D_{Tip}) were selected to evaluate efficiency. Similar notions of motion efficiency are used in established robotic skill evaluation frameworks such as GEARS [47]. Additional motion-level analyses were conducted to support the interpretation of observed performance differences. These metrics were also used to analyze representative subjects from each group, and the detailed results are presented in Section VI-B.

For the analysis of bending-related motion, the instrument was modeled using a constant-curvature assumption. The bending angle was calculated from tendon displacement obtained from rotary motor encoder values, while the rolling angle was derived from the robot manipulator's rotational measurements. Using these angles, the instrument tip position was computed via forward kinematics. Although the actual bending profile deviates slightly from perfect constant curvature, the maximum positional error under this assumption was estimated to be 1.28 mm. A detailed description of the kinematic computation is provided in the supplementary material.

Perceived workload was assessed using the NASA-TLX questionnaire. The questionnaire was administered on Day 2 after subjects completed all trials for each interface, following the bladder mock-up task. Subjects provided comparative ratings between the two interfaces across six subscales: mental demand, physical demand, temporal demand, performance, effort, and frustration. Each score was reported on a scale from 0 to 100, with higher values indicating greater perceived workload (or lower perceived performance for the performance subscale).

Training tasks and earlier trials (1 to 3) across the bladder mock-up task were used to observe learning trends and were excluded from statistical testing. Trials 4 to 6 of the bladder mock-up, which were considered to reflect a stabilized and sustained interaction state, were used for statistical analysis. Normality was assessed using the Shapiro–Wilk test, and no significant deviation from a normal distribution was observed. However, given the relatively small sample size ($n = 10$), the outcome of the normality test should be interpreted cautiously. For this reason, both parametric and nonparametric paired tests were reported, and effect sizes and confidence intervals were considered together in interpreting the results.

Statistical analyses were conducted in MATLAB (MathWorks Inc., Natick, MA, USA) using paired t-tests and Wilcoxon signed-rank tests, together with effect size measures including Cohen's d and Wilcoxon r . Effect size measures were included to characterize the magnitude and practical relevance of observed differences in post-training performance. Cohen's d (0.2/0.5/0.8) and Wilcoxon's r (0.1/0.3/0.5) were interpreted as small, medium, and large effects, indicating subtle, moderate, and pronounced differences, respectively. Both magnitude and direction of effect sizes, together with test statistics, were considered in evaluating the presence and consistency of observed effects. All values are reported with 95% confidence intervals (CI).

VI. RESULTS AND DISCUSSION

A. Performance across learning phases

A clear learning-related performance trend is observed at the individual level, providing intuitive insight into how interface performance evolves with practice (**Fig. 6(a)**). Across the early trials, the stylus interface exhibited lower task completion times and shorter tip-trajectory distances, with smaller inter-subject variability than the finger interface, indicating superior initial task performance. As trials progressed, both perfor-

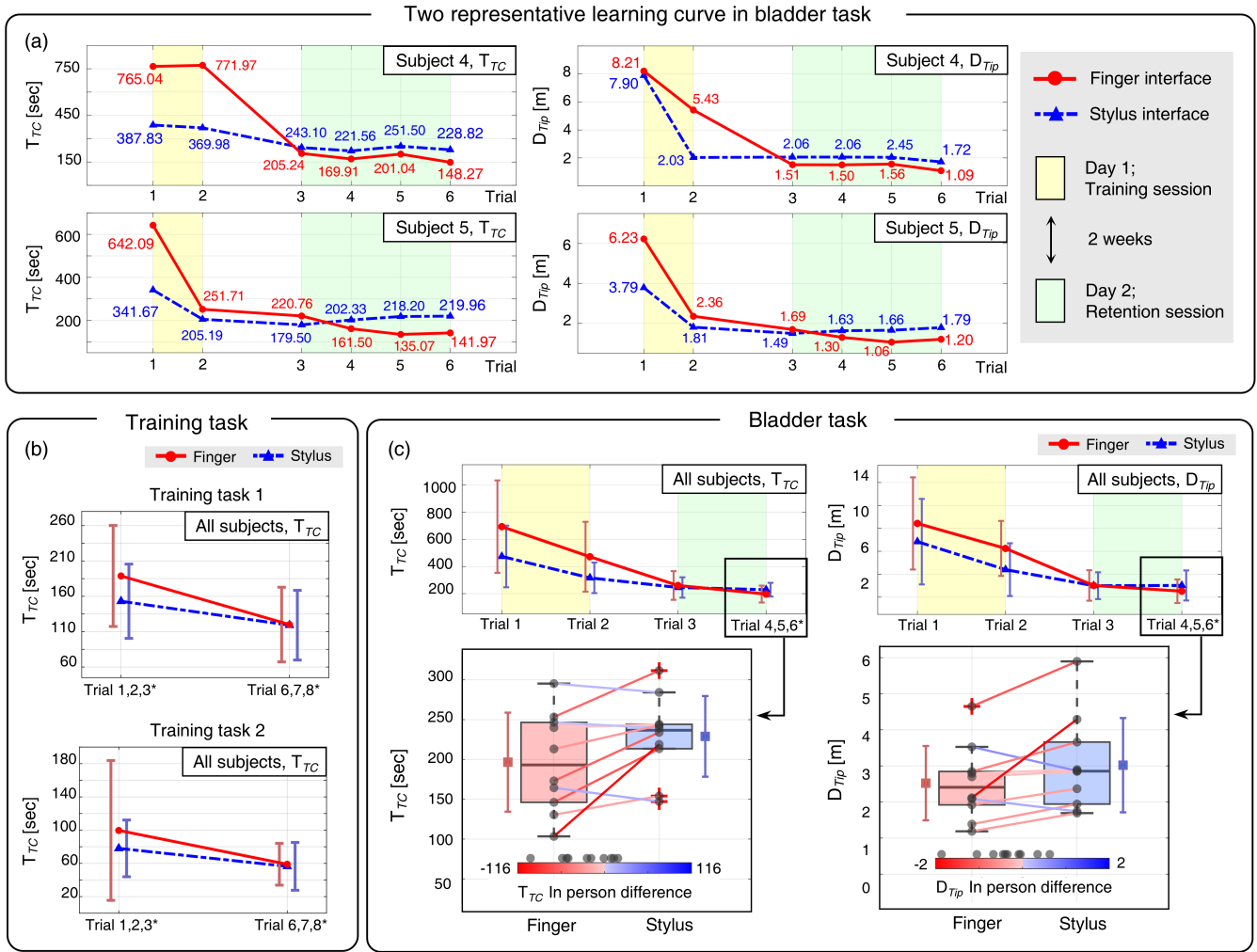


Fig. 6. Comparative analysis of task performance between the finger and stylus interfaces. (a) Representative learning curves from two subjects (top: Subject 4, bottom: Subject 5) in the bladder mock-up task, shown for completion time T_{TC} (left) and tip trajectory distance D_{Tip} (right). (b) Learning curves for Training Task 1 (top) and Training Task 2 (bottom), evaluated using T_{TC} . (c) Learning curves and box plots of averaged performance over Trials 4–6 for the bladder mock-up task, shown for T_{TC} (left) and D_{Tip} (right). Box plots represent subject-averaged values across Trials 4–6, with color intensity indicating relative interface dominance (e.g., red for finger-dominant, black for stylus-dominant).

performance metrics for the finger interface, namely task completion time and total tip trajectory distance, showed larger trial-to-trial reductions, leading to a gradual decrease in performance differences between the two interfaces. In later trials, lower average task completion times and shorter tip trajectories were observed for the finger interface, reflecting improved efficiency with repeated interaction. An additional order/carryover check for the Day 1 crossover session found no significant Interface $\times$ Position interaction for either T_{TC} or D_{Tip} , suggesting no evidence that the crossover structure systematically favored one interface over the other.

The subsequent analysis examines performance trends in greater detail across all subjects. At the beginning of the training session, initial performance with the stylus interface was superior, as reflected by shorter T_{TC} and reduced inter-subject variability, which can be attributed to the familiarity of the interface. As shown in Table I, the stylus interface showed consistently shorter T_{TC} than the finger interface in both Training Task 1 and Training Task 2, based on the aver-

ages from Trials 1 to 3. The standard deviation of T_{TC} was also smaller for the stylus interface in both tasks, indicating more consistent performance across subjects. When examining the average T_{TC} from Trials 6 to 8, the performance gap between the two interfaces appears to have substantially decreased compared to the averages from Trials 1 to 3 (Fig. 6(b)).

This pattern was not limited to the training tasks alone. A similar trend was observed in the bladder mock-up task after training as well (Fig. 6(c)). As the trials progressed, the average differences in both T_{TC} and D_{Tip} between the two interfaces gradually decreased. In the later trials, the finger interface even outperformed the stylus interface on both metrics (Fig. 6(c)). The finger interface was 32.40 seconds faster on average, which corresponds to approximately 14% of the average T_{TC} for the stylus interface and indicates a substantial performance advantage. Similarly, the 0.52-meter reduction in D_{Tip} represents about 17% of the average value for the stylus interface, further reinforcing the improved efficiency. In addition, the finger interface demonstrated a sustained

TABLE I
DESCRIPTIVE STATISTICS OF TASK PERFORMANCE AND SUBJECTIVE WORKLOAD

Metric	Task type	Index	Finger	Stylus	Difference
T_{TC} [sec]	Training Task 1	T1_Trial_1–3	188.80 ± 71.43	153.04 ± 52.50	35.76 ± 72.54
		T1_Trial_6–8	120.09 ± 52.76	118.30 ± 49.28	0.89 ± 50.16
	Training Task 2	T2_Trial_1–3	99.65 ± 84.15	77.82 ± 34.15	21.62 ± 79.63
		T2_Trial_6–8	58.93 ± 25.14	56.42 ± 28.80	2.51 ± 22.89
	Bladder mock-up task	B_Trial_1	693.60 ± 340.76	473.86 ± 227.16	219.74 ± 410.14
		B_Trial_2	471.76 ± 257.27	316.20 ± 112.44	155.56 ± 275.18
		B_Trial_3	259.90 ± 105.71	245.23 ± 75.08	14.67 ± 86.35
		B_Trial_4–6	196.61 ± 62.18	229.00 ± 50.66	-32.40 ± 42.72
D_{Tip} [m]	Bladder mock-up task	B_Trial_1	8.43 ± 4.02	6.84 ± 3.73	1.60 ± 4.26
		B_Trial_2	6.25 ± 2.40	4.44 ± 2.35	-0.19 ± 2.97
		B_Trial_3	3.02 ± 1.33	3.00 ± 1.17	0.03 ± 1.03
		B_Trial_4–6	2.52 ± 1.03	3.02 ± 1.31	-0.52 ± 0.80
NASA-TLX	Bladder mock-up task	Md	46.50 ± 21.99	56.50 ± 13.95	-10.00 ± 18.56
		Pd	47.50 ± 19.18	37.50 ± 25.08	10.00 ± 20.82
		Td	31.00 ± 19.12	42.00 ± 23.24	-11.00 ± 18.53
		Eff	55.50 ± 21.92	53.50 ± 21.86	2.00 ± 22.51
		Perf	47.50 ± 20.58	51.00 ± 17.76	-3.50 ± 25.50
		Frust	40.00 ± 22.11	43.50 ± 18.42	-3.50 ± 14.73

TABLE II
INFERENTIAL STATISTICS OF TASK PERFORMANCE AND SUBJECTIVE WORKLOAD

Index		Paired t-test				Wilcoxon signed rank test			
		<i>P</i> -value	t-stat	Cohen's <i>d</i>	<i>d</i> CI (95%)	<i>P</i> -value	Z	Wilcoxon <i>r</i>	<i>r</i> CI (95%)
B_Trial 4–6	T_{TC}	0.040*	-2.40	-0.76	[-1.46, -0.06]	0.065	-1.89	-0.60	[-0.89, -0.02]
	D_{Tip}	0.081	-1.96	-0.62	[-1.07, 0.08]	0.084	-1.78	-0.56	[-0.88, 0.10]
NASA-TLX	Md	0.123	-1.70	-0.54	[-1.20, 0.12]	0.141	-1.83	-0.58	[-0.89, -0.04]
	Pd	0.163	1.52	0.48	[-0.17, 1.13]	0.219	0.05	0.02	[-0.60, 0.68]
	Td	0.093	-1.88	-0.59	[-1.27, 0.08]	0.109	-2.19	-0.69	[-0.89, -0.27]
	Eff	0.785	0.28	0.09	[-0.53, 0.71]	0.836	-0.25	-0.08	[-0.66, 0.57]
	Perf	0.675	-0.43	-0.14	[-0.76, 0.49]	0.719	-0.87	-0.27	[-0.77, 0.37]
	Frust	0.472	-0.75	-0.24	[-0.87, 0.39]	0.688	-1.68	-0.53	[-0.85, 0.02]

Note. All difference values (as well as associated t-statistics, Z-values, and effect sizes) were computed as “finger interface – stylus interface.” For all metrics, lower values indicate better performance. Therefore, negative values across these measures reflect better outcomes with the finger interface.

learning effect. Trial 3 was conducted on Day 2, which took place two weeks after the initial session (Day 1), and in this session, subjects repeated the same task. For the finger interface, both T_{TC} and D_{Tip} were lower in Trial 3 than in Trial 2.

For T_{TC} , the paired t-test yielded a test statistic of -2.40 ($p = 0.040$), and the Wilcoxon signed-rank test yielded -1.89 ($p = 0.065$) in Table II. Since the paired t-test p -value is below the 0.05 threshold, this result indicates a statistically significant difference. The corresponding effect sizes were -0.76 for Cohen's d and -0.60 for Wilcoxon r , indicating a medium-to-large effect.

The strength of the effect suggests a performance improvement that is not only statistically significant but also likely to be meaningful in practice. The direction and magnitude of the test statistics further support the presence of a consistent trend favoring the finger interface over the stylus interface.

The analysis of D_{Tip} also revealed a consistent trend toward improved performance with the proposed interface. For D_{Tip} , the paired t-test yielded a test statistic of -1.96 ($p = 0.081$), and the Wilcoxon signed-rank test yielded -1.78 ($p = 0.084$).

Although the p -values do not meet the conventional 0.05 threshold, both tests fall below the exploratory 0.10 level and exhibit a consistent trend favoring the finger interface. The corresponding effect sizes were -0.62 for Cohen's d and -0.56 for Wilcoxon r , indicating a medium-to-large effect. The alignment of the test statistics and effect sizes suggests a possible tendency toward more efficient tip motion with the finger interface. However, given the limited sample size and corresponding statistical power of the present study, this result should be interpreted as suggestive rather than conclusive.

B. Motion strategy analysis of representative subjects

A notable factor reflected in the observed performance differences between the two interfaces lies in the control strategies adopted by the subjects, particularly in how their motion patterns suggest different degrees of sensorimotor alignment. Subjects who excelled with the finger interface consistently employed rolling and bending motions while exhibiting reduced reliance on translational movement, indicating effective use of the instrument's articulated degrees of freedom. In contrast, subjects who performed better with the

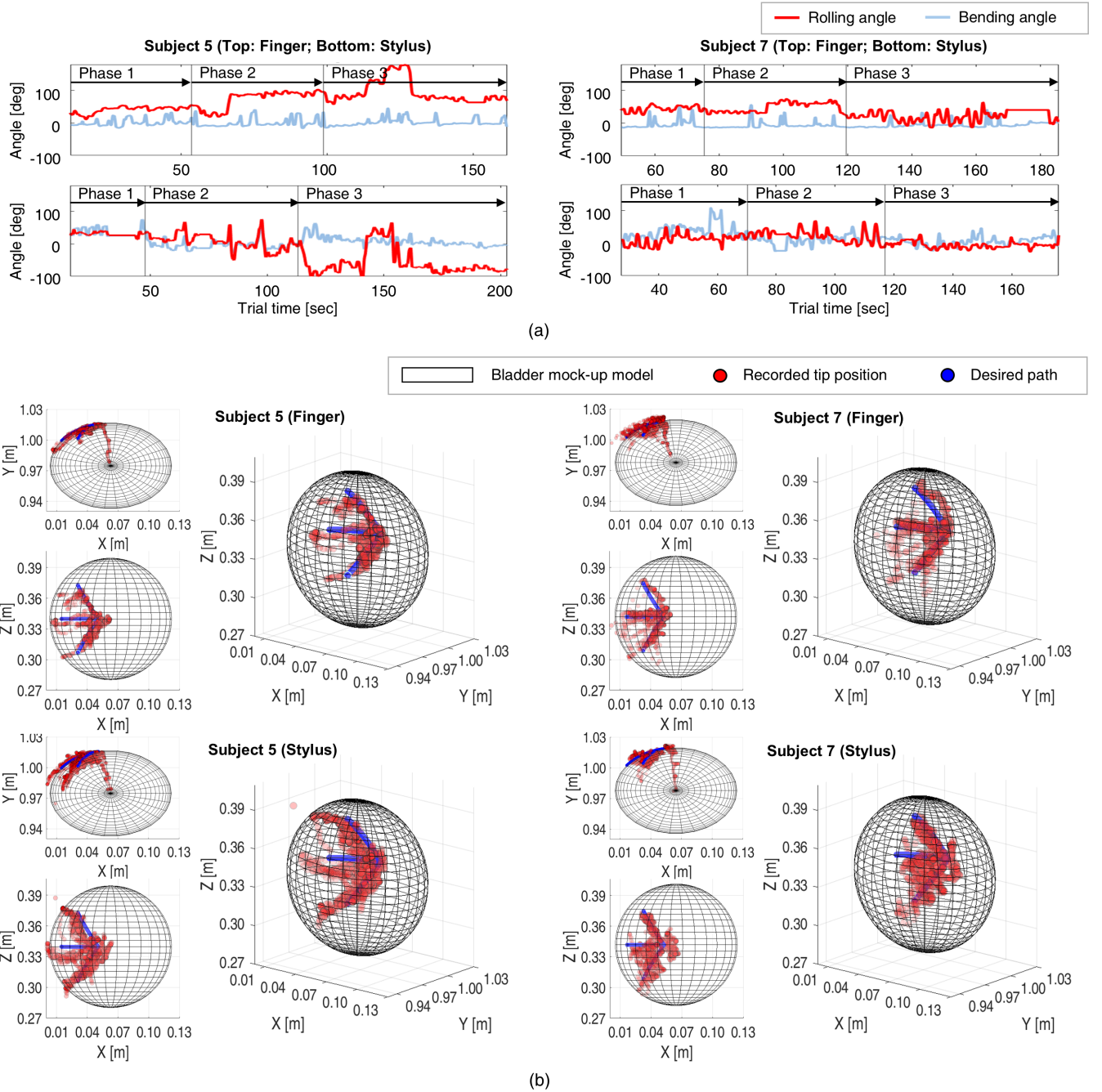


Fig. 7. Representative motion plots of two subjects performing the bladder mock-up task (left: Subject 5, who showed effective use of the finger interface; right: Subject 7, who relied more on translation dominant control): (a) Recorded rolling and bending angles of each interface, (b) Trajectories of each interface shown from three different views.

stylus interface tended to adopt translation-dominant control strategies, showing limited engagement with the instrument's kinematic capabilities. These strategy-level differences provide descriptive support for the proposed embodiment-oriented design perspective, suggesting that interfaces promoting sensorimotor coherence may foster more structured and consistent control behavior during task execution.

Subjects who ranked high in both the T_{TC} and D_{Tip} metrics when using the finger interface consistently adopted control strategies characterized by phase-dependent modulation of rolling and bending motions. As the task progressed from

Phase 1 to Phase 3, their rolling angles increased stepwise, forming a staircase-like pattern in the plots of rolling and bending angles. This phase-dependent modulation was specific to the finger interface and did not carry over to the stylus condition for the same subjects. Subject 5 was a representative example of this behavior (Fig. 7(a)). In contrast, Subject 7—who achieved the best performance in both T_{TC} and D_{Tip} using the stylus interface—showed minimal variation in rolling angles regardless of the interface used. Compared to Subject 5, Subject 7 exhibited fewer adjustments in both rolling and bending angles, indicating a control strategy dominated by

TABLE III
PERFORMANCE COMPARISON ACROSS PHASES OF THE BLADDER TASK

Index		Interface type		Independent t-test				
		Finger	Stylus	P-value	t-stat	Cohen's d	d CI (95%)	
T_{TC}	Phase 1	31.378	39.217	0.197	-1.40	-1.02	[-20.70, 5.02]	
	Phase 2	35.072	43.493	0.014	-3.14	-5.29	[-16.16, -2.46]	
	Phase 3	43.409	65.688	0.018	-2.97	-3.84	[-39.59, -4.96]	
D_{Tip}	Phase 1	0.359	0.469	0.022	-3.67	-1.64	[-0.19, -0.03]	
	Phase 2	0.412	0.461	0.062	-2.56	-1.15	[-0.10, 0.00]	
	Phase 3	0.554	0.725	0.024	-3.53	-1.58	[-0.31, -0.04]	

Note. Task completion time (T_{TC}) and tip trajectory distance (D_{Tip}) across the three phases of the bladder mock-up task (Trials 4–6), based on the top five participants selected separately for each interface.

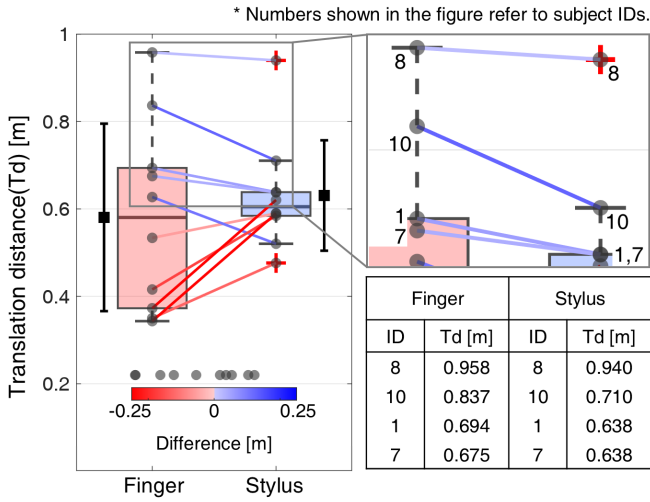


Fig. 8. Translation distance comparison between the finger and stylus interfaces. The four participants with the largest translation distances maintained the same rankings regardless of the interface type.

translational motion with limited engagement of the instrument's bending and rolling motions.

These differences suggest that superior performance with the finger interface was associated with the ability to interpret and exploit the mechanical characteristics of the bendable instrument through coordinated rolling and bending, rather than relying primarily on translation. This distinction in control strategies is further reflected in the recorded tip position trajectories (Fig. 7(b)). By contrast, Subject 7, who relied primarily on translational motion, did not exhibit a consistent trajectory pattern. For Subject 5, examination of the tip positions projected onto the XY and XZ planes reveals trajectories that more closely approximate elliptical curves than those of Subject 7. This spatial pattern suggests that Subject 5 followed the intended curved surface more continuously across the task, reflecting a more consistent and coordinated control strategy.

When subjects were ranked in ascending order based on T_{TC} , those with higher rankings generally exhibited lower translation distances when using the finger interface compared to the stylus interface (Fig. 8). Subject 9, who achieved the fastest T_{TC} , recorded the third-lowest translation distance (0.373 m). Subject 6, ranked second, recorded the lowest translation distance overall, using only 58% of the translation

distance relative to their stylus use. In contrast, subjects with high translation distances (Subjects 1, 7, 8, and 10) consistently relied heavily on translation motion regardless of the interface. Notably, Subject 8, who recorded the highest translation distance, showed the smallest difference between the two interfaces (0.018 m), suggesting that they interacted with both interfaces using similar control strategies. Consistent with this pattern, Subject 7 ranked among the top performers when using the stylus interface.

The observed interface-specific control differences were statistically examined across task phases corresponding to different target positions, focusing on the top five performers for each interface. For each interface, subjects were ranked based on the corresponding performance metrics, and the top five were selected; as a result, some subjects appeared in the top five for both interfaces. Performance differences between interfaces in the bladder mock-up task were then evaluated across target locations by conducting statistical tests on T_{TC} and D_{Tip} for each phase.

For the T_{TC} metric, the p-value for Phase 1 was 0.197, indicating no statistically significant difference (see Table III). In contrast, statistically significant differences were found in Phase 2 and Phase 3, with p-values of 0.014 and 0.018, respectively. This pattern is consistent with the possibility that, whereas motions required for the lower-position target remained relatively compatible with familiar pen-like movements afforded by the stylus interface, the middle and upper target positions demanded motion patterns that were less common in typical stylus use. Under these conditions, the finger interface, which was designed according to embodiment-oriented considerations, may have supported faster task completion. For the D_{Tip} metric, statistically significant differences were observed in Phase 1 and Phase 3, with p-values of 0.022 and 0.024, respectively, while Phase 2 exhibited a marginally significant difference ($p = 0.062$). Collectively, these results suggest that the finger interface tended to produce shorter tip-trajectory distances across task phases, consistent with reduced unnecessary motion and potentially more efficient control.

C. Workload analysis

Temporal demand showed the largest numerical difference among the NASA-TLX subscales between the two interfaces (Fig. 9). However, the statistical tests did not reach conventional significance levels. The p-values were 0.093 for the paired t-test and 0.109 for the Wilcoxon test. Although the corresponding effect sizes (Cohen's $d = -0.59$ and Wilcoxon $r = -0.69$) were moderate, these results should be interpreted cautiously because the statistical tests were not significant. This pattern may suggest a possible tendency toward lower perceived temporal demand with the finger interface, but does not provide conclusive evidence of a difference between the interfaces.

Mental demand showed a similar pattern, with relatively lower p-values (0.123 and 0.141) and moderate effect sizes (-0.54 and -0.58). However, these values likewise did not reach statistical significance. The remaining subscales—effort, performance, and frustration—produced p-values near or

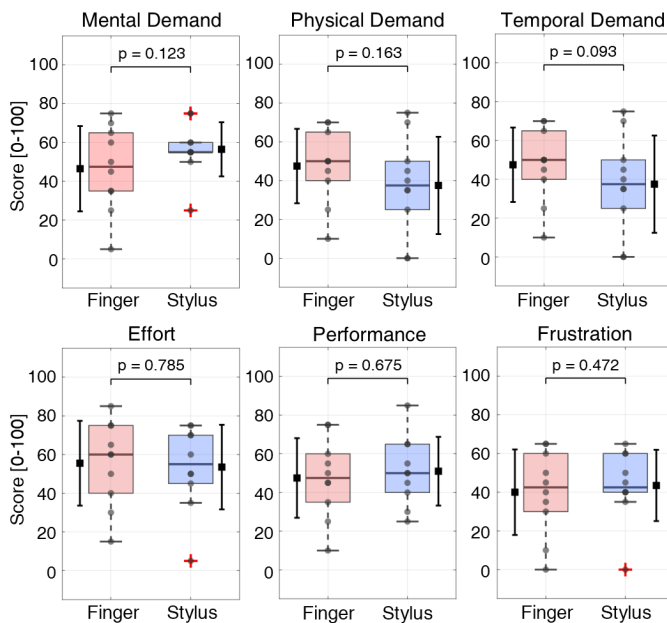


Fig. 9. Comparison of NASA-TLX subscale scores between the finger and stylus interfaces. Lower scores indicate lower perceived workload across all subscales; for the performance subscale, lower ratings indicate higher perceived performance. Annotated p-values are based on paired t-tests, while Wilcoxon signed-rank test results are also provided in Table 2.

above 0.5, indicating no clear evidence of differences between the interfaces. Physical demand yielded p-values of 0.163 and 0.219, and the corresponding effect sizes likewise did not indicate a clear between-interface difference.

Overall, none of the NASA-TLX subscales showed statistically significant differences between the two interfaces. Accordingly, these workload-related findings should be interpreted as suggestive rather than conclusive.

VII. CONCLUSION

A. Summary of Findings

This work addresses the challenge of designing intuitive interfaces for tele-robotic systems whose kinematics depart from those of familiar tools. Rather than treating intuitiveness as an immediate outcome reflected in initial task performance, the results support a view of intuitiveness as something that develops across interaction and learning, shaped by how coherently the operator can organize sensorimotor behavior within the control loop. In this context, the proposed framework provides a structured perspective for relating interface design choices to different phases of interaction, linking strategy selection with their system-level realization through the notion of the Perceived Body and its associated interaction mechanisms, thereby enabling a more systematic design and evaluation of intuitive teleoperation interfaces across learning phases.

The empirical findings from telerobotic surgery setting further illustrate this perspective. While the familiarity-driven stylus interface facilitated more stable and consistent initial performance, the embodiment-oriented finger interface was associated with greater performance improvements over repeated trials, more stable behavior after a temporal gap,

and the emergence of more structured motion strategies that better utilized the robot's kinematic capabilities. These phase-dependent patterns suggest that interface intuitiveness cannot be fully characterized by early usability alone, but is closely tied to how interaction structures support learning and adaptation.

B. Limitations and Future Work

First, the present study compared instantiated interface strategies rather than isolating individual design factors. In this setting, elements such as control mapping, kinematic correspondence, and interface-level mechanical characteristics were inherently coupled in how each strategy was realized for the same follower robot and task. The findings should therefore be interpreted as empirical evidence of the practical usefulness of the proposed strategy-level framework, rather than as the isolated effect of any single interface feature. Future work can further examine how such strategy-level comparisons may be conducted more systematically, including ablation-style studies and additional baselines to more clearly disentangle the roles of mapping, mechanical support, and other interface-level design choices.

Second, system- and interface-level implementation factors may also influence the observed interaction and learning behavior. For example, characteristics such as system update rate and end-to-end latency can affect the responsiveness of the control loop, particularly in tasks requiring faster interaction or higher precision. In addition, the mechanical properties of the interface—including spring stiffness, inertia, and passive physical cues—may contribute to control stability, user strategy, and perceived task difficulty. While these factors were not the primary focus of the present study, they represent important aspects of interface realization that merit more comprehensive investigation in future work, including systematic variation of timing conditions and interface mechanical characteristics.

Finally, the present empirical validation was limited to a single case study involving a distally bendable surgical instrument, a specific task structure, and a relatively small participant group. The findings should therefore be interpreted within this context rather than as directly generalizable across all teleoperation systems. In addition, because the present study was evaluated in a novice population, direct generalization to trained surgeons should be made with caution. Prior familiarity with conventional surgical instruments and established motor strategies may alter the relative balance between initial usability and longer-term adaptation across interface strategies. Broader validation will therefore require follow-up studies across different robot morphologies, task demands, sensory conditions, and user populations, including trained surgeons and settings in which different parts of the robot may become more functionally salient to the operator than in the present case.

VIII. CONFLICT OF INTEREST

Some authors are inventors on patents related to the finger interface design presented in this work. The authors declare that these patents did not affect the scientific conduct or reporting of the study.

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